

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)			The {sequence / order} of the (DNA) bases (is known)		1		1		
	(b)			Any three (x1) from A. Do not need to {locate / isolate / find} the gene / do not need to use gene probes (1) B. No need to {use restriction enzymes to cut up DNA / find functional mRNA} (1) C. (Restriction enzyme) may cut within the gene / extracting the gene may damage it (1) D. No {introns / non coding DNA} present / only {exons / coding DNA} present (1)		3		3		
	(c)			<u>Restriction</u> {endonuclease / enzyme} cuts the DNA (in the plasmid) at a specific {sequence of bases / point / owtte} (1) (DNA) ligase {anneals / joins / binds} the {plasmid / vector} and {gene / DNA / owtte} (1) Award 1 mark for 2 correct enzymes if no credit given for description	2			2		2
	(d)	(i)		Only bacteria which have a <u>plasmid</u> can {survive / grow} / shows that these bacteria have taken up a <u>plasmid</u> / OWTTE		1		1		
		(ii)		(The two colonies that) {did not grow / died / are missing} are the bacteria which are {not resistant / are susceptible} to <u>ampicillin</u> (1) - As they have taken up the {HGH gene / recombinant plasmid / recombinant DNA} (1) - So the <u>ampicillin</u> gene is {not expressed / split / interrupted / owtte} (by HGH gene) (1)		1	2	3		3
		(iii)		That they will produce HGH (when in culture) / that the HGH gene is {functional / can be expressed}			1	1		1
	(e)			2.9 X 10 ¹⁴ 2 marks If incorrect award 1 mark for sight of 10 ⁶ x 10 ⁸ 10 ¹⁴ 2900000000000000		2		2	2	
				Question 3 total	2	8	3	13	2	6